Project Template “Machine Learning in Engineering Applications” (Winter Semester 25/26)

Team Members.

Given Name	Surname	Matriculation No.	Study Course
Shariqa	Moin	1511284	Global Software development
Rashid Hussain	Mohammed	1540260	Global Software development
Revathi	Podaralla	1542884	Data Science

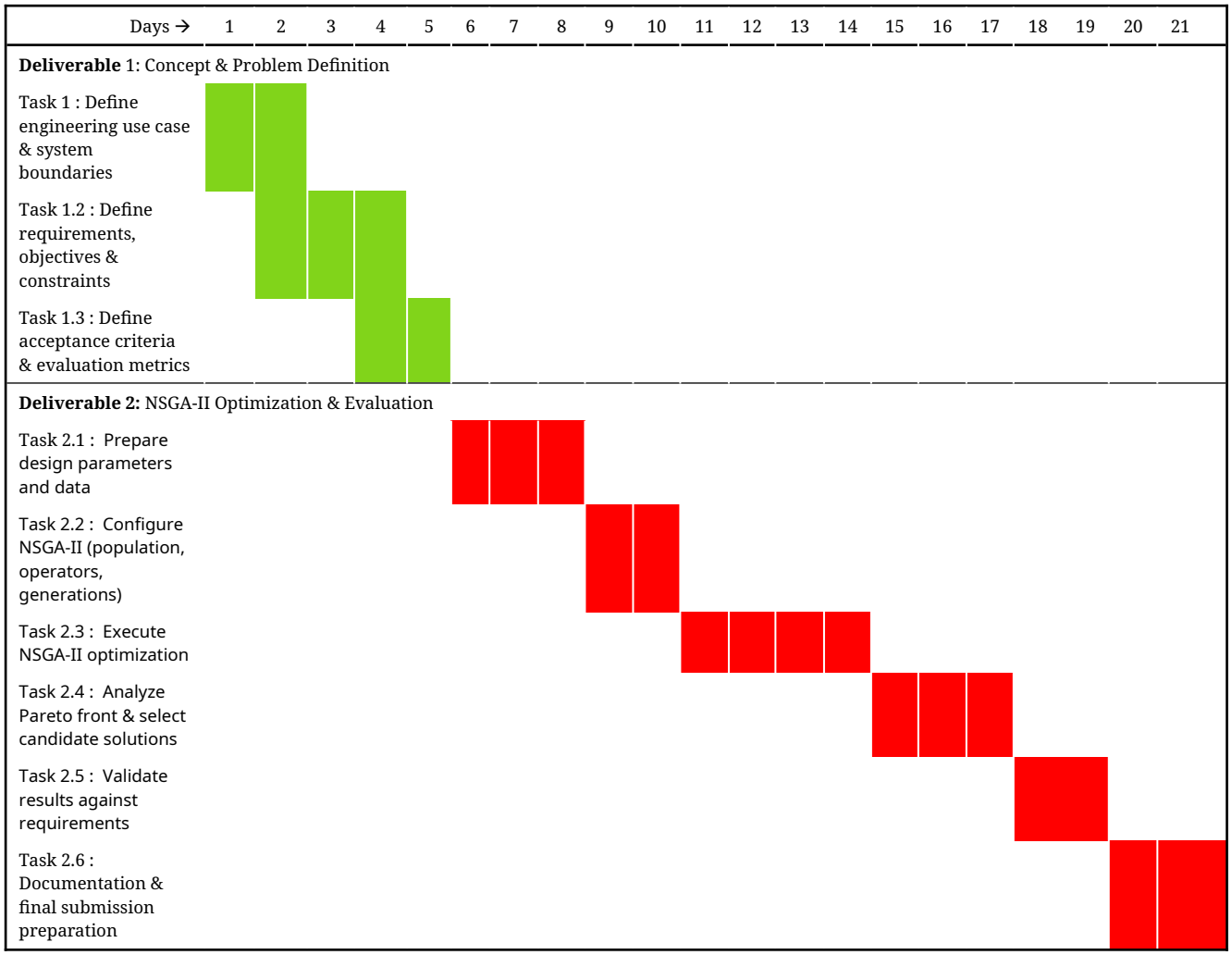
Roles & Responsibilities (RACI-Matrix)

Tasks & Competencies for AI-Based Systems	Project Manager	Systems Engineer	Systems Architect	Requirements	Acceptance	Integration	Risk Manager	Ethics Advisor	Legal Risks Officer	Test Engineer	Data Engineer	Data Scientist	Machine Learning	Maintenance Engineer
Integration of AI procedures in overall system and software architecture	I	A	R	I	C	C	I	C	C	I	I	I	C	I
Assessment of current and future legal risks as well as economic analysis of legal risks	C	I	I	C	I	I	A	C	R	I	I	I	C	I
Anticipation and responding to ethical issues and definition of a code of conduct	A	I	I	I	C	I	C	R	C	I	I	I	I	I
Ensure compliance with data privacy and/or security regulations	C	I	C	C	C	I	C	I	A	I	R	I	I	C
Scout and define suitable machine learning algorithms according to requirements	I	I	I	C	I	I	C	I	C	I	C	A	R	I
Define data acquisition, cleaning and management concept in development and operational phase	I	I	I	C	I	C	I	C	C	C	A	R	C	C
Adapt/define data quality framework as well as attributes for assessment of training data	I	I	I	C	I	I	I	C	I	I	R	A	C	I
Perform acceptance scoring based on operational system concept and user requirements	C	C	I	R	A	I	I	C	C	I	I	I	C	I
Define multi-objective optimization problem (objectives & constraints)	I	C	I	C	I	I	I	I	I	I	C	R	A	I
Configure NSGA-II parameters (population size, generations, operators)	I	C	I	I	I	I	I	I	I	I	I	C	R	I
Execute NSGA-II optimization and generate Pareto front	I	C	I	I	I	I	I	I	I	I	C	C	R	I
Analyze Pareto-optimal solutions and select candidate designs	I	C	I	C	C	I	I	I	I	C	I	R	A	I
Validate optimization results against engineering requirements	I	C	I	C	A	I	I	I	I	R	I	C	I	I

Role Model Justification: The selected roles are derived from the professional AI-system development setup provided in the course material. For this specific project, the team’s main focus is only on the roles critical for a conceptual NSGA-II–based design optimization prototype, ensuring all engineering, data, and management aspects are covered efficiently by the three team members.

Team Member	Assigned Roles	Core Responsibilities
Rashid	• Machine Learning Expert • Data Scientist • Systems Engineer	Responsible for defining the multi-objective optimization problem, configuring and executing the NSGA-II algorithm, defining objective functions and constraints, and ensuring the technical correctness of the engineering optimization approach.
Shariqa	• Project Manager • Requirements Engineer • Acceptance Manager	Responsible for project coordination and scheduling, definition and documentation of system and user requirements, and specification of acceptance criteria for evaluating optimization results.
Revathi Podaralla	• Data Engineer • Test Engineer • Integration Engineer	Responsible for preparation and management of design parameters and data, validation of optimization results, and testing the generated Pareto-optimal solutions against defined requirements.

Gannt-Chart (3 weeks total)



Milestone Descriptions (at least 1 per Team Member)

Abbreviation Gannt-Chart	Short Description Milestone	Acceptance Criteria	Responsible Role (1 role)
M1	Concept & Requirements Defined	The engineering use case, optimization objectives (Fuel vs. Performance), and system constraints are fully documented and approved.	Requirements Engineer
M2	NSGA-II Prototype Operational	The NSGA-II Python implementation is complete, successfully connects to the simulation environment, and generates valid offspring generations.	Machine Learning Expert
M3	Optimization Results Validated	Full optimization experiments are concluded, the Pareto-optimal front is visualized, and solutions meet the defined engineering acceptance criteria.	Test Engineer
M4	Final Submission Ready	All project deliverables (Source Code, Evaluation Report, Presentation) are finalized, reviewed, and packaged for submission.	Project Manager

to be filled by lecturer

Submission Date: Project-ID:

The project proposal was submitted on time and meets
the assessment criteria for examination as project work.

Signature Lecturer Machine Learning